

EXPERIMENT NO. 14

STUDY OF LOGIC GATES

Aim : To identify given IC's and to verify the truth table for each type of logic gate.

Theory: Gate is a logic circuit with one or more input signals, but only one output signal. Gates are digital circuits because the input and output signals are either low (0) or high (1) voltages. A truth table shows all input and output possibilities for a logic circuit. Total number of combinations is 2^n variables the number of inputs. OR, AND and NOT are the basic gates, NAND and NOR are derived gate the OR get has a high output if any input is high. The AND gate provide a high output only when all inputs are high. For Not gate output is always opposite the input. it provides light output when its input is low The NAND gate has a high output when at least one of its inputs is low. The NAND gate has a high output when all inputs are low.

The ICs 7400, 7402, 7404, 7408, 7432 are from standard TTL family. For TTL devices, any voltage from 0 to 0.8 V is considered low input and any voltage from 2 to 5V is consider high input. Any voltage from 0 to 0.4 V is low output and any voltage from 2.4 to 3.9 V is high output. For 7400 series supply voltage rage is 4.75 to 5.25 V over a temperature range of 0°C to 70°C .

Logic gates can be used to build flip-flops, digital counters etc. which can be used calculators, computers and other digital systems.

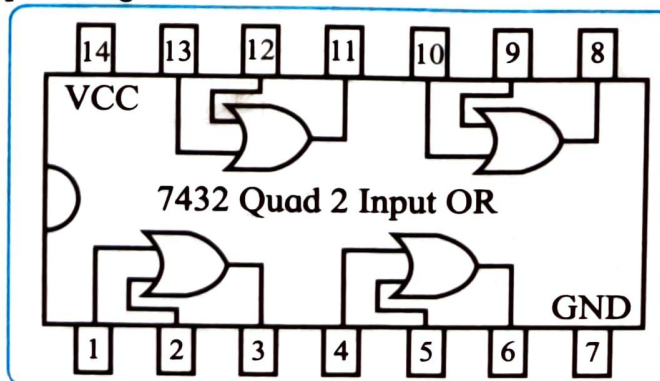
Apparatus : 5V DC power supply, IC block ,probes, digital multimeter. IC7408 (AND Gate), IC 7432 (OR Gate), IC 7402 (NOT Gate), IC 7400 (NAND Gate), IC 7404 (NOT Gate)

Procedure:

1. Mount the IC to be tested on the IC block.
2. Connect the positive of the supply to pin 14 and negative to the 7th pin of the IC.
3. Consider any one of its gate out of four gates.
4. Give different combinations of input from 0 to 1. Here 0 indicates zero voltage and 1 indicate +5V.
5. Connect a voltmeter across the output terminal of IC and the ground terminal to pin no.7.
6. Measure the output voltage, which will be zero for 0 and around +5V for 1 state.
7. Repeat the same procedure to test the other gates.
8. Verify the truth table and observation table for different input combinations.
9. Repeat the above procedure to test the other IC's i.e. 7408, 7404, 7400,7432 etc.

Pin Diagram and observations :

- (1) Quad – dual – Input OR gate IC .



For OR Gate no.1, pin nos. 1 and 2 are inputs and pin no. 3 is output.

IC 7432 (OR GATE)

Logic Equation

$$Y = A + B$$

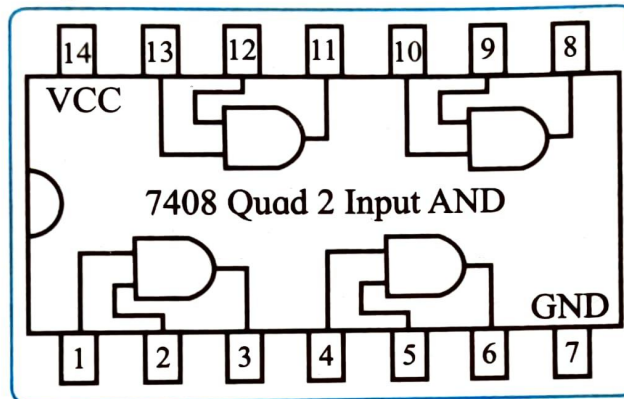
Truth table

INPUT		OUT PUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Observation table

INPUT		OUT PUT
A (V)	B (V)	Y (V)
0	0	0
0	1	1
1	0	1
1	1	1

2) Quad – dual – input AND gate IC



IC 7408 (AND GATE)

Logic Equation

$$Y = AB$$

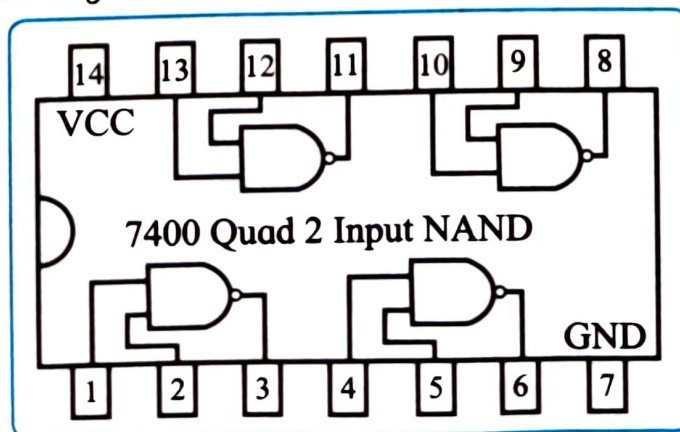
Truth table

INPUT		OUT PUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Observation table

INPUT		OUT PUT
A (V)	B (V)	Y (V)
0	0	0
0	1	1
1	0	1
1	1	1

3) Quad- dual- input NAND gate IC



IC 7400 (NAND GATE)

Logic Equation

$$Y = \overline{A \cdot B}$$

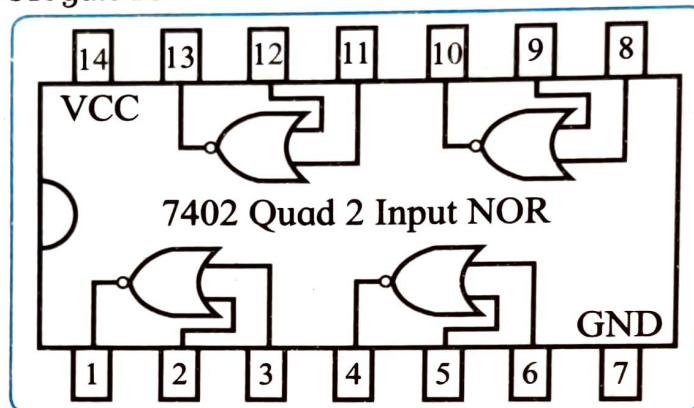
Truth table

INPUT		OUT PUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Observation table

INPUT		OUT PUT
A (V)	B (V)	Y (V)
0	0	0
0	1	1
1	0	1
1	1	1

4) Quad-dual- input NOR gate IC



IC 7402 (NOR GATE)

Logic Equation

$$Y = A + B$$

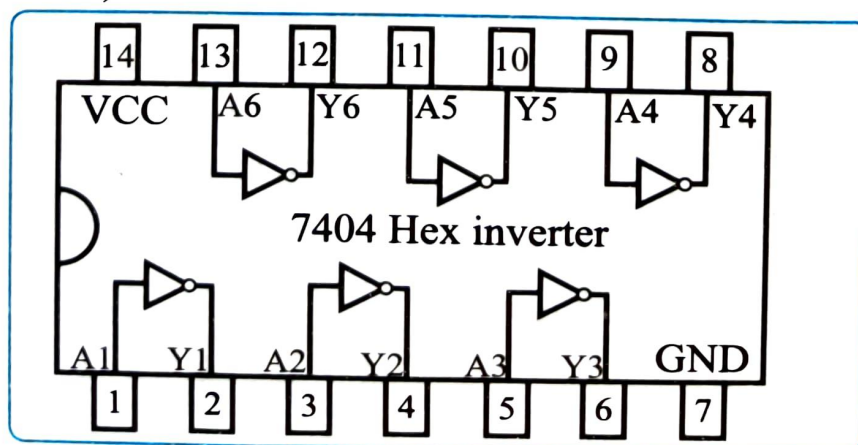
Truth table

INPUT		OUT PUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Observation table

INPUT		OUT PUT
A (V)	B (V)	Y (V)
0	0	0
0	1	1
1	0	1
1	1	1

Hex Inverter (NOT Gate)



IC 7404 (HEX INVERTER) NOT GATE

Logic Equation

$$Y = A$$

Truth table

INPUT		OUT PUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Observation table

INPUT		OUT PUT
A (V)	B (V)	Y (V)
0	0	0
0	1	1
1	0	1
1	1	1

Precautions :

1. Input voltage must be 5 volts.
2. Connections should be joined correctly.
3. Grounding should be proper for input, 0' volt.
4. Connections must be tight.

Result: The different given logic IC's are tested and their truth table is verified.

Questions

1. Explain what are the basic logic elements?

The basic logic elements are the fundamental logic gates used to build digital circuits. The three basic logic gates are AND, OR and NOT gates. All digital systems can be constructed using these logic gates.

2. What are the applications of logic gates?

Logic gates are used in computers, calculators, digital communication systems, microprocessors, microcontrollers, control circuits and other electronic devices for performing logical operations.

3. NAND Gate is combination of which two basic Gates?

A NAND gate is a combination of an AND gate followed by a NOT gate. It gives the complement of the output of an AND gate.

4. NOR Gate is combination of which two basic Gates?

A NOR gate is a combination of an OR gate followed by a NOT gate. It gives the complement of the output of an OR gate.

Remark and sign of teacher :